

Practice Problems: Calculus

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

Exercise 6.2

2. Given the function $y = 5x^2 - 4x$:
 - (a) Find the difference quotient as a function of x and Δx .
 - (b) Find the derivative dy/dx , using the limit definition.
 - (c) Find $f'(2)$ and $f'(3)$.
3. Given the function $y = 5x - 2$:
 - (a) Find the difference quotient $\Delta y/\Delta x$. What type of function is it?
 - (b) Since Δx does not appear in $\Delta y/\Delta x$ in part (a), does it make any difference to the value of $\Delta y/\Delta x$ whether Δx is large or small? Consequently, what is the limit of the difference quotient as Δx approaches zero?